

Right Triangles: Which Method?

Answer Key

High School Geometry

Quick Answers

- | | |
|------------------|----------------------------------|
| 1. 15 cm | 6. 19.05 ft |
| 2. 24 in | 7. 15 ft |
| 3. 9.90 cm | 8. 44.04 m |
| 4. 10.83 m | 9. 33.37 degrees |
| 5. 28.07 degrees | 10. (a) leg length = 16.97 cm, — |
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What is verified

9 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: High School Geometry

HSG-SRT.C.8 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 11 tagged checks.

Common wrong answers

1. *If they got 21: added the two legs instead of using the theorem*
 2. *If they got 27.86: added the squares instead of subtracting, treating the hypotenuse as a leg*
 4. *If they got 14.38: used cosine instead of sine and returned the adjacent side*
 6. *If they got 6.35: multiplied by the tangent instead of dividing by it, shrinking the long leg*
 9. *If they got 28.81: used the inverse tangent although the second known side was the hypotenuse*
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1. Legs 9 cm and 12 cm; find the hypotenuse, in centimetres.

Solution. Two sides and no angle, so the Pythagorean theorem applies:

$$c = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225}.$$

Therefore

$c = 15 \text{ cm}$

2. Hypotenuse 26 in, one leg 10 in; find the other leg, in inches.

Solution. The unknown is a leg, so the known hypotenuse is subtracted from, not added to:

$$b = \sqrt{26^2 - 10^2} = \sqrt{676 - 100} = \sqrt{576}.$$

Therefore

$$b = 24 \text{ in}$$

3. Isosceles right triangle with both legs 7 cm; find the hypotenuse, in centimetres.

Solution. Equal legs means a 45–45–90 triangle, whose hypotenuse is a leg times $\sqrt{2}$ (Pythagoras gives the same: $\sqrt{7^2 + 7^2} = 7\sqrt{2}$):

$$c = 7\sqrt{2} = 9.8994\dots$$

Therefore

$$c = 9.90 \text{ cm}$$

4. Hypotenuse 18 m, acute angle 37° ; find the opposite side, in metres.

Solution. An angle that is not 30° , 45° or 60° means trigonometry. Opposite with hypotenuse is sine:

$$\text{opp} = 18 \sin 37^\circ = 10.8326\dots$$

Therefore

$$10.83 \text{ m}$$

5. Legs 8 and 15; find the acute angle opposite the leg of length 8, in degrees.

Solution. Two legs and an unknown angle: the pairing is opposite with adjacent, so use the inverse tangent.

$$A = \tan^{-1}\left(\frac{8}{15}\right) = 28.0724\dots^\circ$$

Therefore

$$28.07^\circ$$

6. One acute angle 30° with the opposite leg 11 ft; find the other leg, in feet.

Solution. This is a 30–60–90 triangle: the leg opposite 30° is the short one, and the other leg is $\sqrt{3}$ times it. Using the tangent gives the same number, with the unknown underneath:

$$b = \frac{11}{\tan 30^\circ} = 11\sqrt{3} = 19.0525\dots$$

Therefore

$$b = 19.05 \text{ ft}$$

7. A 17 ft ladder with its foot 8 ft from a wall; how far up the wall does it reach, in feet?

Solution. The ladder is the hypotenuse and the ground distance is a leg; no angle is given, so Pythagoras:

$$a = \sqrt{17^2 - 8^2} = \sqrt{289 - 64} = \sqrt{225}.$$

Therefore

$$15 \text{ ft}$$

8. A path of 32 m to the foot of a tower, with the top seen at 54° ; find the tower's height, in metres.

Solution. The 32 m path is adjacent to the 54° angle and the height is opposite it, so tangent:

$$a = 32 \tan 54^\circ = 44.0442 \dots$$

Therefore

44.04 m

9. A 40 ft cable reaching 22 ft up a mast; find the angle the cable makes with the ground, in degrees.

Solution. The 22 ft height is opposite the angle and the 40 ft cable is the hypotenuse, so the pairing is sine and the angle is the inverse sine:

$$A = \sin^{-1}\left(\frac{22}{40}\right) = 33.3670 \dots^\circ$$

Therefore

33.37°

10. A square tile cut along a 24 cm diagonal. (a) Find the length of each leg of one resulting right triangle, in centimetres. (b) Explain how the special right-triangle relationship gives the same answer and why it is quicker.

Solution (a). The diagonal of a square makes two 45–45–90 triangles with the diagonal as hypotenuse, so each leg is opposite a 45° angle:

$$\text{leg} = 24 \sin 45^\circ = 16.9705 \dots$$

Therefore

16.97 cm

Solution (b) — instructor-judged. A complete response names the triangle as 45–45–90, states that the hypotenuse is a leg times $\sqrt{2}$, and rearranges to $\text{leg} = 24/\sqrt{2} = 12\sqrt{2}$, the same 16.97 cm. It should add that the special-triangle route is exact and needs no calculator, while the sine route reaches the same value only after evaluating a trigonometric function. Award full credit only when both the alternative method and a reason for preferring it appear.